

RAKSHIT NAIDU NEMAKALLU

[Email](#) ♦ [LinkedIn](#) ♦ [Website](#) ♦ [Google Scholar](#)

OBJECTIVE

My research interests hover around topics in Ethical Machine Learning (ML), Trustworthy/Responsible Artificial Intelligence (AI), and AI for societal good. I'm interested in creating applications that have a direct impact on society through my research.

EDUCATION

Doctor of Philosophy (Ph.D.) in Machine Learning, Georgia Institute of Technology
2023 - (Expected 2028)

Master of Science (M.Sc.) in Information Technology (Privacy Engineering), Carnegie Mellon University
2021 - 2022
Selected Courses: [Ethics in Machine Learning](#), [Foundations of Privacy](#), Privacy Policy, Law and Technology (PPLT) and [ML with Large Datasets](#)

Bachelor of Technology (B.Tech.) in Computer Science and Engineering, Manipal Institute of Technology
2017 - 2021
Minor in Computational Mathematics
Selected Courses: Computational Linear Algebra, Distributed and Cloud Computing, Graph Theory and Matrices.

EXPERIENCE

Graduate Assistant May 2025 - July 2025
Georgia Institute of Technology *Atlanta, GA, USA*

- Selected for the [GT OSPO Virtual Summer Internship Program 2025](#).
- Working with the [IBM Granite](#) team on creating a recipe for evaluating Generics in LLMs (Project Mentor – [Dr. Alessandra Pascale](#) and Supervising Mentor – [Dr. Susan Malaika](#)).
- Outcome – Published [IBM Developer](#) article : [Evaluating generic phrases using Granite models](#).

Graduate Research Intern Apr 2023 - July 2023
Carnegie Mellon University *Pittsburgh, PA, USA*

- Worked with [Prof. Hoda Heidari](#) as a Research Assistant in the Machine Learning Department (MLD) at CMU.
- My responsibilities entailed of collecting, assimilating, and analyzing both qualitative and quantitative data from prior academic publications, with the goal of creating a tool that offers a pipeline-aware view of Fairness for Machine Learning to researchers and practitioners.
- Outcome – Accepted as [oral talk at ACM EAAMO'23](#) and [NeurIPS'23 tutorial](#) ([Toward Operationalizing Pipeline-aware ML Fairness](#)).

Visiting Research Scholar Jun 2022 - Aug 2022
Syracuse University *Syracuse, NY, USA*

- Worked with [Prof. Ferdinando Fioretto](#) on topics related to Differential Privacy and Fairness in AI.
- Outcome – Accepted as [Spotlight Talk at NeurIPS'22](#) ([Pruning has a disparate impact on model accuracy](#)).

Application Engineering Intern Jan 2021 - Jul 2021
BlackRock *Gurugram, India (Remote)*

- Part of the Client-End Fund Reporting Team. Improved test coverage on FRED (Factsheet Reporting Engine and Distribution) and fixed code issues, blockers and bugs.
- Received an honourable mention for our internal hackathon project on “BlackRock’s Cultural Heatmap” which provides a forum for both employees (to assess their mental and cultural well-being) and managers (to maintain a cultural pulse throughout the organization).

PUBLICATIONS & PROJECTS

- “Fairness-as-a-defense”: How fair models defend against distribution inference attacks** [Link \(soon!\)](#)
Rakshit Naidu
- Personalized Differential Privacy for Ridge Regression** [Link](#)
*Krishna Acharya, Franziska Boenisch, **Rakshit Naidu**, Juba Ziani*
(Accepted at [Naval Research Logistics \(NRL\)](#) (Q1 journal))
- Are Chatbots Ready for Privacy-Sensitive Applications? An Investigation into Input Regurgitation and Prompt-Induced Sanitization** [Link](#)
*Aman Priyanshu, Supriti Vijay, Ayush Kumar, **Rakshit Naidu**, Fatemehsadat Mireshghallah*
(Preprint)
- Toward Operationalizing Pipeline-aware ML Fairness: A Research Agenda for Developing Practical Guidelines and Tools** [Link](#)
*Emily Black, **Rakshit Naidu**, Rayid Ghani, Kit Rodolfa, Daniel Ho, Hoda Heidari*
(Accepted at [ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization \(EAAMO\)](#), 2023 (archival))
★ Oral Presentation (~18% acceptance rate)
★ [NeurIPS’23 tutorial \(AI Governance & Accountability for Machine Learning: Existing Tools, Ongoing Efforts & Future Directions\)](#)
- Can Causal (or Counterfactual) Representations benefit from Quantum Computing?** [Link](#)
***Rakshit Naidu**, Daniel Justice*
(Accepted as an extended abstract at [Algorithmic Fairness through the Lens of Causality and Privacy \(AFCI\)](#) workshop at NeurIPS’22)
- Pruning has a disparate impact on model accuracy** [Link](#)
*Cuong Tran, Ferdinando Fioretto, Jung-Eun Kim, **Rakshit Naidu***
(Accepted at [NeurIPS’22](#))
★ Spotlight Lightning Talk (~3% acceptance rate)
(★) Nomination for Best Paper Award
- Fair Context-Aware Privacy Threat Modelling** [Link](#)
*Saswat Das, **Rakshit Naidu***
(Presented at [Privacy Threat Modeling \(PTM\)](#) workshop at USENIX-SOUPS’22)
- Can Causal (and Counterfactual) Reasoning improve Privacy Threat Modelling?** [Link](#)
***Rakshit Naidu**, Navid Kagalwalla*
(Presented at [Privacy Threat Modeling \(PTM\)](#) workshop at USENIX-SOUPS’22)
- Efficient Hyperparameter Optimization for Differentially Private Deep Learning** [Link](#)
*Aman Priyanshu, **Rakshit Naidu**, Fatemehsadat Mireshghallah, Mohammad Malekzadeh*
(Accepted at [PPML](#) workshop at ACM CCS’21 and as a poster at [IEEE-S&P’21](#))
- Privacy Enabled Financial Text Classification using Differential Privacy and Federated Learning** [Link](#)
Priyam Basu, Tiasa Singha Roy*, **Rakshit Naidu**, Zumrut Muftuoglu*
(Accepted at [Economics and Natural Language Processing \(ECONLP\)](#) workshop at EMNLP’21)
- Benchmarking Differential Privacy and Federated Learning for BERT models** [Link](#)
Priyam Basu, Tiasa Singha Roy*, **Rakshit Naidu**, Zumrut Muftuoglu, Sahib Singh, Fatemehsadat Mireshghallah*
(Accepted at [Machine Learning for Data: Automated Creation, Privacy, Bias \(ML4Data\)](#) workshop at ICML’21)
- Towards Quantifying Carbon Emissions of Differentially Private Machine Learning** [Link](#)
Rakshit Naidu, Harshita Diddee*, Ajinkya Mulay*, Aleti Vardhan, Krithika Ramesh, Ahmed Zamzam*
(Accepted at [Socially Responsible Machine Learning \(SRML\)](#) workshop at ICML’21)
- DP-SGD vs PATE: Which Has Less Disparate Impact on Model Accuracy?** [Link](#)
Archit Uniyal, **Rakshit Naidu***, Sasikanth Kotti, Patrik Joslin Kenfack, Sahib Singh, Fatemehsadat Mireshghallah,*

Andrew Trask

(Accepted at [ML4Data](#) workshop at ICML'21 and [PPML](#) workshop at ACM CCS'21. And also as a poster at [IEEE-SEIP'21](#))

FedPerf: A Practitioners' Guide to Performance of Federated Learning Algorithms

[Publication](#)

Ajinkya Mulay*, Baye Gaspard*, **Rakshit Naidu***, Santiago Gonzalez-Toral*, Vineeth S*, Tushar Semwal*, Ayush Manish Agrawal

(Accepted for publication at [PMLR](#))

When Differential Privacy Meets Interpretability: A Case Study

[Link](#) — [Poster](#)

Rakshit Naidu*, Aman Priyanshu*, Aadith Kumar, Sasikanth Kotti, Haofan Wang, Fatemehsadat Mireshghallah

(Accepted as extended abstract at [Responsible Computer Vision \(RCV\)](#) workshop at CVPR'21; full paper accepted at [Privacy-Preserving Machine Learning \(PPML\)](#) workshop at ACM CCS'21)

Improved variants of Score-CAM via Smoothing and Integrating

[Poster](#)

Rakshit Naidu, Soumya Snigdha Kundu, Ankita Ghosh, Yash Maurya, Shamanth R Nayak K, Joy Michael, Haofan Wang

(Accepted as extended abstract at [Responsible Computer Vision \(RCV\)](#) workshop at CVPR'21)

FedPandemic: A Cross-Device Federated Learning Approach Towards Elementary Prognosis of Diseases During a Pandemic

[Link](#)

Aman Priyanshu, **Rakshit Naidu**

(Accepted at [Distributed and Private Machine Learning \(DPML\)](#) and [Machine Learning for Preventing and Combating Pandemics \(MLPCP\)](#) workshops at ICLR'21)

SS-CAM: Smoothed Score-CAM for sharper visual feature localization

[Link](#)

We introduce Smoothing to the Score-CAM algorithm, which is a state-of-the-art CAM algorithm. Smoothing allows us to capture more features of the focused object in the image, which leads to better visually attributed results.

IS-CAM: Integrated Score-CAM for axiomatic-based explanations

[Link](#)

We borrow the idea of integration from “IntegratedGrad” and combine it with Score-CAM to conduct faithfulness evaluations. IS-CAM performs better than SS-CAM and Score-CAM in terms of faithfulness evaluations, considering the VGG-16 as our baseline model.

TeleVital: Enhancing the quality of contactless health assessment

[Paper](#) — [News](#)

Our team came 2nd in a pan-Indian hackathon called #CODE19 and won \$5000 for this solution to detect vitals from the webcam itself, thereby promoting remote diagnosis during COVID-19. I worked on the Respiratory rate calculations via webcam and was responsible for documenting the entire project for presenting at the hackathon.

AWARDS & PROFESSIONAL SERVICES

- Served as Graduate Teaching Assistant (GTA) for the following courses:
 - (Spring 2025, Fall 2025, Spring 2026) Deterministic Optimization (ISYE 6669) – Online.
 - (Fall 2024) Computational Data Analysis (CSE/ISYE 6740).
 - (Fall 2022 Mini-2) [Quantum Circuit Mappings \(17-620\)](#)
 - (Fall 2022 Mini-1) [Programming Quantum Computers \(17617-A1\)](#)
 - (Spring 2022) [Quantum Computing Theory and Lab \(11-860\)](#)
- Served as a Reviewer for:
 - AISTATS'25
 - NeurIPS'23 (+Ethics Reviewer)
 - [PPAI](#)-25, 24, 23, 22 @ AAAI (Program Committee Member)
 - [AFCP](#) @ NeurIPS'24, 23, 22.

- [GenLaw](#) @ ICML’23
- [XAI](#) @ NeurIPS’23
- [SaTML](#)’23 (Conference grant \$1000)
- [ACM EAAMO](#)’23 (Conference grant \$750)
- Our team (Emily, Hoda, Kit, Rayid, Daniel and I) will be presenting a NeurIPS’23 tutorial called “[AI Governance & Accountability for Machine Learning: Existing Tools, Ongoing Efforts & Future Directions](#)”, based on our EAAMO’23 paper! Congratulations to everyone involved!
- Recipient of the Thomas H. Johnson Fellowship award for the academic year 2023-24.
- Talk at Comcast Cybersecurity team (headquartered in Philadelphia, PA) on “Context-Aware Privacy Threat Modeling”. The same talk was also delivered at the [Privacy Threat Modeling workshop at SOUPS 2022](#).
- TEDxMAHE Countdown 2020 Speaker on *Federated Learning for Climate Change*. [Event Link](#) — [Talk](#)
- Manipal Conclave 2020 Student Speaker on *Privacy for ML*. [Memento](#)
- Poster Presented at PyCon India 2019 on *Secure and Private AI with PySyft*. [Poster](#)
Volunteered at PyCon India 2020.
- Used to maintain a few cryptography-related packages in Julia such as [Shamir.jl](#) and [DiffPrivacy.jl](#).

EXTRA-CURRICULAR ACTIVITIES

- I speak 7 languages!
- I love Geography and I have the superpower to name any country’s capital (or vice versa) in the world!
- Peaked Top 5 across India on Geoguessr (see [here](#))!
- Elected as the “Social Media/PR coordinator” for the Georgia Tech Badminton Club (GTBC) for the academic year 2024-25.
- Played for the CMU Badminton team in the Fall 2022 Eastern Collegiate MidAtlantic Conf (Badminton Tournament Regionals) held at University of Maryland, College Park in October 2022.
- Finished a full marathon (42 km) at Manipal Marathon 2020 with a timing of 6 hours and 33 minutes. [Certificate](#)